

Solution Requirements

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Weather Data Input	Accepts ten weather parameters through a Flask web form for flood prediction.	High
2	Input Validation	Validates all required fields and numeric values before processing.	High
3	Data Preprocessing	Applies the trained StandardScaler before prediction.	High
4	Flood Prediction	Uses the trained Random Forest model to classify Flood Risk or No Flood Risk.	High
5	Result Presentation	Displays prediction with a clear status	High

		message on the result page.	
6	Model Loading	Loads the saved model and scaler using Joblib during application startup.	Medium
7	Error Handling	Shows meaningful messages for invalid or incomplete inputs.	Medium
8	Modular Architecture	Separates preprocessing, prediction, configuration and model loading for easy maintenance.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should be generated quickly after user input.	< 2 seconds
2	Scalability	Handle multiple prediction requests without reloading the model.	Model loaded once
3	Security & Data Privacy	Processes only weather parameters; no	No PII stored

		personal data is stored.	
4	Reliability & Availability	Handles invalid inputs without crashing.	Graceful error handling
5	Usability & Accessibility	Responsive interface with labels, placeholders and validation.	User-friendly interface
6	Maintainability	Supports easy model and dataset updates through modular code.	Easy future updates